

Synthesizing Strategic Insight: Evaluating Fine-Tuning and Structured Reasoning for Professional-Grade Sports Commentary

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1 Introduction

Formula one is a sport deeply grounded in research and development, that generates billions in revenue each year. With almost a billion international fans watching each race, it is a shame that the sport hasn't become more accessible. In addition, F1 teams are progressively becoming more data-driven and reliant on their strategy and development of their teams and cars. This paper proposes an approach that drives strategic insight via commentary suitable for the consumer and for the teams aiming to acquire assistance with data interpretation. From this paper, consumers of Formula 1 with consumer level hardware should be able to access and use our solution, and teams may be able to find unseen insight promoting improved strategy.

Previous solutions and or proposed technology have not existed for this work, and most likely is soon to be considered with many of the Formula teams creating strategic partnerships with world leaders in artificial intelligence. This paper aims to provide direction towards a valuable asset that may become standard within not only Formula 1 teams' strategy, however additionally adds to the existing corpus of research regarding autonomous sports commentary generation by incentivizing synthesized insight. We believe that via a learned intelligent mapping from live telemetry and previous commentary, a fast consumer grade tool can provide useful intuition.

This paper illustrates the next step in Formula 1 commentary, and a different approach to other sports autonomous sports commentary generation. This is done with the model being built for synthesized insight, in lieu of providing a mere verbatim narration of the visual elements. Furthermore, this approach is optimized to perform on consumer level hardware, ensuring accessibility for consumers, and less powerful potentially embedded applications.

2 Related Work

Automatic Sports Commentary Generation

- [1]
- [2]
- [3]

The research regarding intelligent autonomous sports commentary is extremely sparse, as often regarded within each source’s related works. Formula one is not short of research; however, within this domain, the work is within infancy. Our work, which involves applying such commentary to Formula 1, is a challenge only made possible via similar works done within baseball and basketball. The work undertaken by Lee on [2] provides an insight on how we can make our commentary more insightful from an automated narrative.

DPO vs. RLHF

- [4]
- [5]
- [6]

A prime motivation within commentary been to provide listeners with insightful, most importantly entertaining dialogue. In previous contexts regarding large language models and the broader scope of natural language tasks, we have seen advancements in tailoring responses to match human preferences. A breakthrough within this realm has come from the explorations undergone by Ouyang [5]. Although the benefits from introducing RLHF can greatly improve the commentary generated by an approach found by Rafailov [4] would reap the benefits of such implementation without requiring as much overhead. Both are notable potential additions upon our proposed methodology; however, with the added cost of compute or dataset additions, we are unable to implement such without risk of straying from the affordance of being consumer feasible and achievable within a month of our development. With more time, we would aim to implement DPO into our system.

Proper Dataset Creation

- [7]

In contrast to existing datasets and achievements from similar works such as Zhang [7], Formula 1 constantly makes regulation overhauls on average every 6-8 years. This creates an issue in dataset creation, as we are unable to use a significantly large set of previous years’ telemetry data. Additionally, we were unable to ethically source visual data, which resulted in an absence of visual media within our dataset. In the creation of our dataset, we used strategies learned from Zhang to create a suitable dataset based upon the most up to date regulations within our targeted sport. We relied on historical telemetry data from OpenF1 and publicly available ground truth commentary.

3 Methodology

In this work, we propose a model-driven pipeline for generating Formula 1 commentary from structured telemetry data. The system converts raw race telemetry into a structured representation, extracts semantically meaningful events, and generates commentary using a fine-tuned language model. The design prioritizes reproducibility, low-latency inference, and feasibility on consumer-grade hardware.

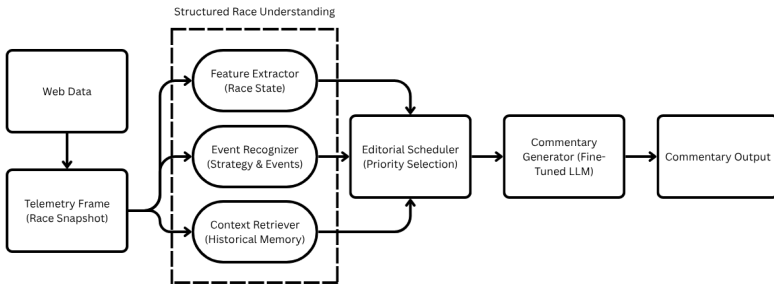


Fig. 1. Proposed telemetry-to-narrative architecture. Structured race-state features, event recognition, and contextual retrieval are combined through an editorial scheduler to generate grounded commentary using a fine-tuned language model.

As illustrated in Figure 1, the system follows a structured pipeline that transforms telemetry data into natural language commentary through feature extraction, event recognition, prioritization, and generation.

3.1 Telemetry Representation

The input to the system consists of historical Formula 1 telemetry data obtained from FastF1. The data is organized into discrete telemetry frames, where each frame represents a timestamped snapshot of the race.

Each telemetry frame contains per-driver state, including position, lap number, gap to car ahead, tire compound, tire age, and DRS status, as well as global race information such as flag conditions and race control events. These frames serve as the fundamental unit of reasoning and are processed sequentially to simulate a live race environment.

To ensure reproducibility and consistent temporal ordering, telemetry data is processed through a lightweight replay mechanism that emits frames in strict timestamp order.

3.2 Feature Extraction

Raw telemetry values are transformed into a structured race-state representation through a feature extraction module. This module computes interpretable features that capture the current dynamics of the race.

The extracted features include gap to car ahead and behind, relative pace differences, tire degradation indicators, DRS availability, pit window relevance, and race control signals. This step defines a structured representation of the race state that enables downstream reasoning over meaningful signals rather than raw numerical inputs.

3.3 Event and Strategy Recognition

On top of the extracted features, the system identifies higher-level semantic events using rule-based and heuristic logic. This step converts numerical signals into interpretable race situations.

Detected events include overtake opportunities, pit strategy scenarios such as undercut or overcut potential, tire degradation phases, safety car or yellow flag conditions, and active battles between drivers. This abstraction enables the system to focus on strategic insight rather than direct observation of telemetry values. This approach is consistent with prior work in automated sports commentary, which emphasizes structured event understanding before language generation [1, 3].

3.4 Editorial Scheduling

Since multiple candidate events may occur simultaneously, the system employs an editorial scheduling mechanism to determine which event should be narrated.

Each candidate event is assigned a priority score based on contextual importance, urgency, race phase, and relevance to ongoing storylines. High-priority events such as lead battles or safety-critical situations are favored over lower-impact developments.

Only the highest-priority event is selected for commentary generation at each time step, while lower-priority events are delayed or suppressed. In addition, a lightweight narrative control mechanism reduces repetition and ensures progression of commentary over time.

3.5 Context Retrieval

To enhance commentary with contextual information, the system includes an optional retrieval component based on a local vector database.

When triggered, the system retrieves relevant historical or strategic context, such as prior battles between drivers or recent race developments. Retrieval is gated and only applied to high-priority events in order to maintain low latency and avoid unnecessary overhead.

3.6 Commentary Generation Model

The final commentary is generated using a fine tuned language model. We employ a Gemma based model adapted through supervised fine tuning with efficient parameter adaptation techniques.

The fine tuning process follows the supervised fine tuning paradigm, where the model is trained on paired input output examples consisting of structured race state representations and corresponding professional style commentary. During training, the model learns to map structured telemetry derived features and event descriptors to coherent natural language outputs by minimizing the cross entropy loss between predicted and reference tokens. This approach enables the model to align its generation behavior with domain specific linguistic patterns and commentary style [6].

The model takes as input the structured race state representation, the recognized event type, and optional retrieved context. It produces short form commentary describing the current race situation.

The model is optimized to generate commentary that is concise, contextually relevant, and strategically informative, while remaining suitable for real time or near real time usage on consumer hardware. The fine tuning approach follows standard practices for adapting language models to domain specific tasks [6].

A lightweight grounding mechanism is applied during generation to ensure that produced statements remain consistent with the underlying race state and derived features.

3.7 Dataset Construction

The training dataset is constructed by aligning telemetry data with professional race commentary. Historical telemetry sequences are segmented into event-centered windows, which are then paired with corresponding commentary transcripts.

Each training example consists of a structured input containing observed facts and derived features, and a target output consisting of the corresponding commentary. This process enables supervised learning of the mapping from telemetry to natural language. The dataset construction strategy is inspired by prior work on sports commentary datasets [7].

3.8 System Implementation

The system is implemented as a lightweight local pipeline to ensure reproducibility and accessibility. Key implementation choices include replay-based telemetry processing, in-memory state tracking, and quantized model inference using local frameworks.

To support evaluation and analysis, the system maintains structured logging of telemetry inputs, generated outputs, and latency measurements. These logs enable reproducibility, debugging, and quantitative comparison across different model configurations.

4 Experiments

4.1 Latency and User Feasability

As our solution is required to run simultaneously with a live race, latency is exceptionally important. What is not fair about this undergoing analysis on this metric is that the hardware in which our system executes can quite obviously boost our performance and lower our latency; this is why we will be aiming for common consumer hardware for testing. The system will run on three different hardware setups, modeling what we believe to be the most common consumer hardware as of 2026. As the tech world has been largely buying mac mini's to run AI solutions, we will be undergoing testing on a 2022 MacBook Pro M1 chip with 16 GB of unified memory, a M4 MacBook Air with 16 GB of unified memory, and additionally a Linux laptop with an Nvidia RTX 4060 to gauge higher-level hardware. We will be aiming for a latency from cognition (when the telemetry packet has been recieved) to final output (when the model has began generation) of less than 1.5 seconds across all devices.

4.2 Our Solution vs. Frontier Models

Additionally, we will be conducting A/B testing between our solution and Frontier models, including Google Gemini 3 Fast, Claude Sonnet 4.6, and ChatGPT 5.4. We will be using Evaluation scores from BERTscore, ground truth accuracy for hallucination analysis, and a human survey-based preference for subjective quality. The goal of this evaluation is to capture quantitative data of our open source solution against frontier alternatives. With this, we will be able to see how we can tune our solution to improve matching the expectations of consumers, and possibly see if it makes more sense to use frontier models within our system.

References

1. Kim, B.J., Choi, Y.S.: Automatic baseball commentary generation using deep learning. In: Proceedings of the 35th Annual ACM Symposium on Applied Computing. (2020) 1056–1065
2. Greg, L., Vadim, B., Elliot, L.: Sports commentary recommendation system (scores): Machine learning for automated narrative. Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment **8** (2012) 32–37
3. Jung, S., Kim, H., Park, H., Choi, A.: Integrated ai system for real-time sports broadcasting: Player behavior, game event recognition, and generative ai commentary in basketball games. Applied Sciences **15**(3) (2025)
4. Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C.D., Finn, C.: Direct preference optimization: Your language model is secretly a reward model (2024)
5. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C.L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askill, A., Welinder, P., Christiano, P., Leike, J., Lowe, R.: Training language models to follow instructions with human feedback (2022)
6. Pareja, A., Nayak, N.S., Wang, H., Killamsetty, K., Sudalairaj, S., Zhao, W., Han, S., Bhandwaladar, A., Xu, G., Xu, K., Han, L., Inglis, L., Srivastava, A.: Unveiling the secret recipe: A guide for supervised fine-tuning small llms (2024)
7. Zhang, B., Gao, J., Yuan, Y.: A descriptive basketball highlight dataset for automatic commentary generation. In: Proceedings of the 32nd ACM International Conference on Multimedia. MM '24, New York, NY, USA, Association for Computing Machinery (2024) 10316–10325